

1 There are three processes. Process A uses 10s CPU successively, 5s device a and 5s CPU, 10s device b and 10s CPU respectively. Process B uses device a for 10s, CPU for 10s, and device b for 5s, CPU for 5s, and device b for 10s. Process C uses device b for 5 s, CPU for 10s, and device a for 5s, CPU for 10s, and device b for 10s. When the two processes occupy CPU or device at the same time, the first requesting process is first served. three processes run at the same time.

Question:

What is the CPU utilization? (Draw the Gantt chart)

2 Suppose that there are five processes in existence $P=\{P0,P1,P2,P3,P4\}$, there are resources A, B, C in the system. Suppose in a moment has the following states:

	Allocation				Max				Available		
	A	B	C		A	B	C		A	B	C
P0	0	0	3		0	0	4		2	4	0
P1	1	0	0		1	7	5				
P2	1	3	5		2	3	5				
P3	0	0	2		0	6	4				
P4	0	0	1		0	6	5				

Question:

- 1) Is the current system in a safe state?
- 2) If the Available resources in the system are (0,5,3), is the system secure? If the system is in a secure state, give a secure sequence; If the system is in an unsafe state, briefly explain why.